

Technical note
Underwriting loan using behavior patterns in rotating savings associations.

(For internal discussion only)

1. The information problem

The central constraint to extending credit in financially thin environments is information asymmetry. When bureau histories are incomplete or absent, lenders cannot directly observe repayment type. Rotating savings and credit associations (ROSCAs) literature can be useful in designing a metric to assess creditworthiness because they partially solve this information problem through repeated interaction, peer observability, and embedded discipline, according to science-based research. From a theoretical view, they can be understood as repeated-game institutions with endogenous enforcement (Besley et al. 1993). Empirically, ROSCAs often function as commitment devices that help members maintain disciplined periodic payments (Gugerty, 2007). Governance quality, sanctions, and social connectedness shape enforcement and sustainability over time (McNabb et al. 2019). Calomiris and Rajaraman (1998) and Klonner (2003) indicates that in bidding ROSCAs, the bid discount behaves like an implicit interest rate and reveals liquidity stress and shocks.

A key bridge to formal underwriting is the insight that publicly observable characteristics inside these informal institutions can predict creditworthiness. The characteristics include:

- + **Payment discipline and persistence** capture time-consistent commitment (Gugerty 2007).
- + **Allocation order and post-payout slippage** encode incentive compatibility and moral hazard after receiving the pot (Besley et al. 1993).
- + **Governance and enforcement strength** reflect heterogeneity in rules, sanctions, and connectedness (McNabb et al. 2019).
- + **Discount level and volatility** reveal liquidity stress (Calomiris and Rajaraman 1998 and Klonner 2003).
- + **Social capital** (repeat interaction, centrality, endorsements) raises reputational stakes and strengthen informal enforcement (Schreiner and Nagarajan 2000).

Schreiner and Nagarajan (2000) provide direct evidence from Gambian ASCRAs/ROSCAs that cheap, observable reliability markers contain predictive signal for credit screening. This is precisely the operational premise of this note: turn socially embedded observables into an interpretable underwriting signal, then calibrate it into a formal probability of default once outcomes accumulate.

2. Formalization

2.1. Target

The long-run objective is to estimate $PD@3M$: the probability that a loan reaches d_{pd60} within 90 days after disbursement. We define the calibration label:

$def3Mi = I\{DPD60 \text{ occurs on or before day } 90 \text{ post-disbursement.}\}$

All predictors must be computed using only information available at the underwriting decision timestamp $\$T_i\$$. This “decision-time discipline” prevents leakage and ensures deployability.

Payout timing matters too. Payout timing affects (i) the availability of payout-related predictors at underwriting and (ii) the borrower’s effective liquidity position. When payout occurs after the credit decision, payout-related features are not observable at τ_i and therefore cannot be used in the underwriting score. Moreover, payout timing may influence repayment capacity and, in some product configurations, may reduce effective exposure through netting mechanisms, affecting observed default risk. Accordingly, we need to consider an underwriting-time payout indicator:

$$tdec_i = I\{adate_i \leq T_i\}$$

so that payout-related features are used only when $tdec_i = 1$ (this account for structural differences in risk associated with pot receipt).

2.2. Data dictionary

The table below summarizes the minimum variables needed to create the metric combining the five dimensions identified. All timestamps are required to respect the decision-time cutoff.

Table1. Data dictionary

Field	Level	Type	Definition
def3m	loan	{0,1}	Def3M label (defined above)
dpd60	loan	{0,1}	dpd60 flag
aord	Member/cycle		allocation order
adate	member	date	allocation date (pot receipt)
tdec	decision	datetime	underwriting decision timestamp (feature cutoff)
mid	member	string	member identifier
gid	group	string	group identifier
cid	cycle	string	cycle identifier
mdate	meeting	date	monthly meeting date (if applicable)
due	meeting	number	amount due
paid	meeting	number	amount paid
dlate	meeting	int	days late (0 if on time)
ont	meeting	{0,1}	on-time flag: 1 if dlate= 0
n	cycle	int	group size
aord	member/cycle	int	allocation order position (1,...,n)
rtype	group	enum	type of tontine: random, bidding, ...
rules	group	{0,1}	rules exist
san6	member/group	int	Count of sanctions in last 6 meetings
santype	meeting	enum	type of sanctions: fee, fine, ...
sanamt	meeting	number	monetary amount of sanction sure (if applicable)
sure_n	member	int	number of sureties
sure_type	member	enum	type of sureties: none, individual, pooled, ...
sure_str	member	enum	stregth of sureties: none, weak, strong
bid flag	attempt	{0,1}	bid attempt made
disc	attempt	number	bidding discount (%) recorded per attempt
rep	member	{0,1}	repeat participation in same group across cycles
cent	member	{0,1}	centrality above group median
end_f	member	{0,1}	foreman endorsement (clean) end s
end_s	member	{0,1}	senior endorsement (clean)
loanid	loan	string	loan identifier
tdisb	loan	datetime	disbursement timestamp
ten	loan	int	tenor (in days)
freq	loan	enum	disbursement frequency

2.3. The model

The design objective is to remain interpretable while reflecting the non-linear nature of behavioral risk. **The model design supposes a monthly setup.** Early history is disproportionately informative relative to incremental history, tail volatility, and clustered slippage matter more than small average differences. We therefore use smooth, monotone non-linear maps—sigmoid, exponential decay, and

log transforms—so that the score saturates where it should and becomes sharply sensitive where risk concentrates.

2.3.1. Payment discipline (pdis)

Let K_i be the number of eligible monthly meetings prior to T_i (standard track: $K_i \geq 6$). Index meetings $m = 1 \dots K_i$ (oldest $\rightarrow$ newest). We define time-decay weights:

$$w_{\{im\}} = a^{K_i - m}, 0 < a < 1$$

with a conservative initial setting $a = 0.80$ for monthly cadence (sensitivity-test to do during calibration).

From meeting-level primitives, we derive a time-decayed on-time rate (otr) and average lateness (al).

$$otr_i = \frac{1}{W_i} \sum_{m=1}^{K_i} w_{\{im\}} \cdot ont_{\{im\}}$$

$$al_i = \frac{1}{W_i} \sum_{m=1}^{K_i} w_{\{im\}} \cdot \max(0, dlate_{\{im\}})$$

with: $W_i = \sum_{m=1}^{K_i} w_{\{im\}}$.

Let ls_i denote the maximum number of consecutive late meetings within the eligible window:

$$ls_i = \max_{1 \leq a \leq b \leq K_i} \{b - a + 1 : ont_{\{im\}} = 0 \text{ for all } m \in [a, b]\}$$

Let rc_i denote a recovery term:

$$rc_i = \frac{1}{W_i} \sum_{m=1}^{K_i} w_{\{im\}} \cdot cur_{\{im\}}, \quad cur_{\{im\}} = I\{0 < dlate_{\{im\}} < 7\}$$

Each otr, al, ls, rc capture specific discipline feature, we can derive a score for each by introducing non-linearity:

$$S^{\{otr\}}_i = 18 \cdot \text{sigmoid}(k_{otr} \cdot (otr_i - c_{otr}))$$

$$S^{\{al\}}_i = 6 \cdot \exp(-a_{al} \cdot \text{logonep}(al_i))$$

$$S^{\{ls\}}_i = 8 \cdot \exp(-a_{ls} \cdot \max(0, ls_i - 1))$$

$$S^{\{rc\}}_i = 7 \cdot \text{sigmoid}(k_{rc} \cdot (rc_i - c_{rc}))$$

with: $\text{sigmoid}(x) = \frac{1}{1 + \exp\{-x\}}$, $\text{logonep}(x) = \ln(1 + x)$

c_{otr} , c_{rc} denotes a centering threshold and k_{otr} , k_{rc} governs the slope of the sigmoid transformation. Parameters a_{al} , a_{ls} control the penalization intensity for average lateness, and late streaks respectively. These parameters would be defined with partners in initial phase and calibrated throughout the development.

Parameter	Role	Suggested Range	Rec. Start	Rationale / Practical Note
c_otr	Center of on-time sigmoid	0.75 – 0.90	0.85	Pivot where “acceptable discipline” sits; 85% on-time is a realistic but demanding norm.
k_otr	Slope of on-time sigmoid	8 – 20	12	Ensures noticeable but not cliff-like change around c_{otr} ; below 8 is too flat, above 20 is too jumpy.
a_al	Penalty intensity (Avg Latency)	0.4 – 1.2	0.7	With $\log(1 + al)$, this gives a moderate penalty for 3–7 days late and a stronger penalty for 10+ days without the score “zeroing out” instantly.
a_ls	Penalty intensity (Streaks)	0.4 – 1.0	0.6	Since $ls_i - 1$ is used, this only triggers at 2+ misses. A 0.6 value makes a 3-month streak very costly without being a total “kill-switch.”
c_rc	Center of recovery sigmoid	0.50 – 0.80	0.7	Requires a solid run of “quick fixes” (lateness < 7 days) before full recovery; avoids instant forgiveness.

k_{rc}	Slope of recovery sigmoid	6 – 15	10	Gives a smooth but visible uplift as \$rc\$ improves; avoids binary “off/on” behavior.
\$a	Time-decay factor	0.75 – 0.90	0.8	Effective memory of ~4–6 meetings; recent behavior is the primary driver, but history still carries weight.

We can compute a payment discipline score – normalize to 35 points contribution:

$$S^{\{pdis\}}_i = \min(35, \frac{1}{39}(S^{\{otr\}}_i + S^{\{al\}}_i + S^{\{ls\}}_i + S^{\{rc\}}_i))$$

2.3.2. Order and post-payout incentives (ordr)

To account for the heterogeneous risk profiles and opportunity costs associated with different payout positions, we introduce an **Order-Based Incentive** component. This term serves two functions: it compensates late-cycle participants for their "lender" status and enforces post-payout discipline through a conditional penalty.

Let define allocation bucket as $O_i = \{\text{early, mid, late}\}$ using terciles of a_{ord}/n , and derive base multipliers as:

$$b(\text{late}) = 1, b(\text{mid})=0.6, \text{ and } b(\text{early})=0.3$$

Let define post-payout slippage $slip_i$ as in indicator equal to 1 if a new late streak of length ≥ 2 begins after pot allocation and before T_i ($t_{date}=1$)

We can compute a post-payout incentives score as :

$$S^{\{ordr\}}_i = 15 b(O_i) * (1 - a_{slip} * slip_i).$$

a_{slip} can be understood here as **post-payout enforcement weight** for post-payout delinquency. It dictates what percentage of the *Order Bonus* is forfeited if a borrower violates the trust of the group after they have already taken the capital. The selection of this coefficient requires a balance between two competing economic objectives: **maximum deterrence** and **rehabilitation Incentive**.

1. The Recommended Value: $\pi_{slip}=0.80$

A value of **0.80** is recommended for standard ROSCA environments. This signifies that 80% of the potential "Order Bonus" is stripped immediately upon the detection of a strategic slippage ($slip_i=1$).

- **The Deterrence Effect:** For a late-bucket participant, this represents a loss of 12 points out of 15. This "heavy hand" signals to the group that the breach of trust is the most significant risk to the collective.
- **The Residual Stake:** By leaving 20% of the bonus intact, the model avoids the "**Bankruptcy Effect**." In behavioral economics, if a penalty is so severe that a user's score becomes irrecoverable (hitting zero), the user loses all incentive to continue participating or "curing" their debt.

References

- Besley, T., Coate, S., & Loury, G. C. (1993). The economics of rotating savings and credit associations. *American Economic Review*, 83 (4), 792–810.
- Calomiris, C. W., & Rajaraman, I. (1998). The role of ROSCAs: Lumpy durables or event insurance? *Journal of Development Economics*, 56 (1), 207–216.
- Gugerty, M. K. (2007). You can't save alone: Commitment in rotating savings and credit associations in kenya. *Economic Development and Cultural Change*, 55 (2), 251–282. <https://doi.org/10.1086/508716>

- Klonner, S. (2003). Rotating savings and credit associations when participants are risk averse. *International Economic Review*, 44 (3), 979–1005.
- McNabb, K., LeMay-Boucher, P., & Bonan, J. (2019). Enforcement problems in ROSCAs: Evidence from benin. *European Journal of Development Research*. <https://doi.org/10.1057/s41287-019-00215-5>
- Riley, R. D., Ensor, J., Snell, K. I. E., Harrell, F. E., Martin, G. P., Reitsma, J. B., Moons, K. G. M., Collins, G. S., & van Smeden, M. (2020). Calculating the sample size required for developing a clinical prediction model. *BMJ*, 368, m441. <https://doi.org/10.1136/bmj.m441>
- Schreiner, M., & Nagarajan, G. (2000). *Predicting creditworthiness with publicly observable characteristics: Evidence from ASCRAs and ROSCAs in the gambia* (tech. rep.) Microfinance Risk Management, L.L.C. <https://www.microfinance.com/English/Papers/Gambia RoSCA ASCRA.pdf>
- Vittinghoff, E., & McCulloch, C. E. (2006). Relaxing the rule of ten events per variable in logistic and Cox regression. *American Journal of Epidemiology*, 165 (6), 710–718. <https://doi.org/10.1093/aje/kwj095>